MILITARY SPECIFICATION

DETONATOR, STAB, PA 523
LOADING, ASSEMBLING, AND PACKING

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery for the loading and assembling of Detonator Stab, PA523.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116 -	Preservation, Methods of.
MIL-A-2550 -	Ammunition and Special Weapons, General Specifications for.
MIL-I-48078-	Ammunition Standard Quality Assurance Provisions, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U S Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-650 - Explosive: Sampling, Inspection, and Testing .
MIL-STD-1235- Continuous Sampling Procedures and Table for Inspection by Attributes.

(Copies of specfications, standards, drawings , publications , and other Government documents required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer .)

2. 1.2 Other Government documents , drawings , and publications. The following other Government documents , drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7548099 - Box, Packing, Ammunition for Detonators.
7548103 - Box, Set-up, Packing, Ammunition, Outer for Boosters , Detonators, or Primers, Assembly, Details, Packing and Marking .
7548104 - Box, Set-up, Packing, Ammunition, Inner for Boosters , Detonators and Primers. Assembly Detailed Packing and Marking.
9358084 - Detonator , Stab, PA523 Loading Assembly.

PUBLICATIONS

49 CFR 100-199 Code of Federal Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents , US Government Printing Office, Washington, DC 20402. Orders should specify “49 CFR 100-199 (latest revision)”.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however , shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings, specifications and standards.

3.2 Loading assembly. The loading assembly shall comply with all requirements specified on Drawing (Dwg.) 9358084, all associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Moisture content of lead azide. The moisture content of lead azide used in loading the detonator shall not exceed 0.50 percent when tested as specified in 4.5.1.

3.4 Moisture content of primer mixture. The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.3.

3.5 Functioning.

3.5.1 Sensitivity. The detonator shall function on the first attempt and shall cause perforation of a 0.1345 inch thick lead disc, to a diameter of not less than 0.3125 inch when tested as specified in 4.5.5.1.

3.5.2 Waterproofness . The detonator shall comply with the requirements of 3.5.1 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours at a temperature of 70°F plus or minus 10°, when tested as specified in 4.5.5.2.

3.6 Binder-lubricant content of RDX. The total binder-lubricant, calcium resinate and graphite content of the RDX shall be determined for each lot at the time of loading as specified in 4.5.2.

3.7 Primer mixture composition. The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading as specified in 4.5.4.

3.8 Heat seal test of sealed bag. Sealed barrier bag shall be tested as specified in 4.5.6.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of cracks, gouges, burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required) , corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly, nor shall they be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Resporisibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below and Table I.

Part Description	Drawings	Quantity
PA523, Detonator, Stab	9358084	340

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The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3. Unless otherwise directed by the Contracting Officer, loaded components shall not be submitted to a Proving Ground for any test until the above assemblies submitted in accordance with this paragraph have been accepted.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. If any items fail to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3.3 Rejection. MIL-A-48078 shall apply with the exception that the first article sample shall be rejected if five (5) detonators fail either functioning/waterproof test as specified.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-D-70603(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	PA523, Detonator, Stab				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	PA523, Detonator, Stab (Dwg. 9358084)				
	Examination for Defects	315		3.2	4.4.2.1/4.4.2.2
	Sensitivity test	315-2-3	1/	3.5.1	4.5.5.1
	Waterproofness test	25-0-1		3.5.2	4.5.5.2
	Moisture content lead azide			3.3	4.5.1
	Moisture content of primer mixture			3.4	4.5.3
	Binder-lubricant content of RDX			3.6	4.5.2
	Primer mixture composition			3.7	4.5.4
	Heat seal test of sealed bag			3.8	4.5.6

NOTES: 1/. If one or two defectives occur, a second sample of 315 detonators shall be selected and tested. If in the combined first and second sample, a total of four (4) or more defectives occur, the lot shall be rejected.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality Conformance Inspection.

4. 4.1 Inspect ion lot format ion. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Metal parts assemblies of one lot inter fix number from one manufacturer .

b. Primers of one lot inter fix number from one manufacturer . (See 4.4.5)

c. Lead azide of one lot interfix number from one manufacturer .

d. RDX of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70603(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9358084
4.4.2.1	PA523, Detonator, Stab, prior to loading and closing				NEXT HIGHER ASSEMBLY 8833783
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Interior coating incomplete or inadequate		100%	3.2	Visual
Major	None defined				
Minor					
201	Workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70603(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	PA523, Detonator, Stab Assembly		9358084
			NEXT HIGHER ASSEMBLY 8833783

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split or cracked		100%		Visual
Major					
101	Total length		0.40%	3.2	Gage
102	Diameter		0.40%	3.2	Gage
103	Disc loose		0.40%	3.2	Manual
104	Color of disc improper		0.40%	3.2	Visual
105	Not crimped full 360°		0.40%	3.2	Visual
106	Disc punctured or torn		0.40%	3.2	Visual
107	Moisture content lead azide		0.40%	3.3	4.5.1
108	Moisture content of primer mixture		0.40%	3.4	4.5.3
109	Sensitivity		0.40%	3.5.1	4.5.5.1
110	Waterproofness		0.40%	3.5.2	4.5.5.2
111	Binder-lubricant content of RDX		0.40%	3.6	4.5.2
112	Primer mixture composition		0.40%	3.7	4.5.4
Minor					
201	Workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70603(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Box, set-up, inner		7548104
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Box damaged		0.40%	3.2	Visual
102	Number of detonators incorrect		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Any packing material missing		0.65%	3.2	Visual
203	Workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70603(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Box, set-up, outer				7548103
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Number of inner set-up boxes		0.40%	3.2	Visual
102	Sealing strip torn, badly wrinkled incomplete or otherwise fails to seal the box completely		0.40%	3.2	Visual
Minor					
201	Box torn, cut or punctured		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Contents move when box is shaken		0.65%	3.2	Visual
204	Workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70603(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Sealed barrier bag				7541803
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Seal missing, incomplete or incorrectly located		0.40%	3.2	Visual
102	Bag punctured, torn or cut		0.40%	3.2	Visual
103	Heat seal test of sealed bag		0.40%	3.8	4.5.6
Minor					
201	Workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70603(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Box, packing, ammunition				7548099
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Board broken or split		0.40%	3.2	Visual
102	Strapping missing, broken or loose		0.40%	3.2	Visual
Minor					
201	Marking misleading, unidentifiable		0.65%	3.2	Visual
202	Strapping improperly assembled		0.65%	3.2	Visual
203	Contents move when box is tilted		0.65%	3.2	Visual/Manual
204	Workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraph (See 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Methods of inspection.

4.5.1 Moisture content of lead azide. The moisture content of the lead azide shall be determined in accordance with MIL-STD-650 Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluorothylene. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of lead azide shall be rejected. If loading has begun, action on the unloaded azide shall be as specified above and all detonators loaded with questionable lead azide shall be rejected.

4.5.2 Binder-lubricant content of RDX. A sample of 2 ounces of the RDX and binder-lubricant mixture shall be selected and tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3. If the total binder-lubricant, calcium resinate or graphite is in excess of the percentages specified on the applicable drawing and if loading has not begun, the lot of RDX shall be rejected until proper corrective action has been accomplished as verified by performing the test for binder-lubricant content as specified. If loading has begun, action on the unloaded RDX shall be as specified above and all detonators loaded with the questionable RDX shall be rejected.

4.5.2.1 Calcium resinate. An accurately weighed portion of 5 grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliter (ml.) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20 ml. portion of chloroform to insure complete removal of the calcium resinate. The residue shall be aspirated until the odor of chloroform is no longer detectable; the crucible and contents shall be placed in a drying oven at 95-105°C. for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying and weighing with individual 20 *ml.*

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portions of chloroform until the lose in weight on subsequent extractions total less than 0.003 grams. The loss in weight of the crucible plus contents shall be calculated to percent calcium resinate in the sample on a moisture-free basis as follows:

$$\text{Percentage of Calcium Resinate} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible plus contents.

w = weight of sample

4.5.2.2 Graphite. The crucible and contents remaining after the removal of calcium resinate shall be washed with a minimum of three 20 ml. portions of acetone, each portion maintained in contact with the residue for approximately 1 minute with continuous stirring before applying suction, until the subsequent test for RDX in the acetone washings is negative. The crucible and contents shall be washed with a minimum of 3 ml. portions of acetone, each portion maintained in contact with the residue for approximately 1 minute with continuous stirring before applying suction, until the following qualitative test for RDX in the acetone washing is negative . The acetone filtrate shall be transferred to a 25 ml. beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable: the crucible and contents shall be placed in a drying oven at 95-105°C for 1/2 hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture free basis as follows:

$$\text{Percentage of Graphite} = \frac{100A}{W}$$

Where:

A = increase in weight of filtering crucible.

w = weight of sample.

4.5.2.3 Total binder-lubricant. The percentage of total binder-lubricant in the sample shall be calculated by adding the percentage of calcium resinate and graphite found.

4.5.3 Moisture content of primer mixture. The moisture content of the primer mixture shall be determined in accordance with

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MIL-STD-650 method 101.5 except, use a one gram sample. If the moisture content is in excess of the percentage specified and if loading has not begun, the batch or batches of the primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.5.4 Composition of primer mixture. The composition of the primer mixture shall be determined by performing the tests as specified in 4.5.4.1 - 4.5.4.6. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing and if loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, detonators loaded with the questionable mixture shall be rejected.

4.5.4.1 Barium nitrate. An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml. capacity. Three ml. of lead azide saturated distilled water shall be added at a temperature of 50°C plus or minus 20 C. agitated for 1 minute, and if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3 ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60°C plus or minus 5°C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of-barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{M\ A-B}{1-.01M}$$

Where:

A = weight of crucible and sample.

B = weight of the crucible and sample after extractions with lead azide saturated water.

c = moisture content of sample obtained from 4.3.3 in percent.

4.5.4.2 Basic lead styphnate (when applicable). The residue obtained in the determination of barium nitrate shall be extracted with six ml. portions of saturated ammonium acetate solution at about 25°C (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact

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with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml. aliquot of this solution shall be transferred to a 50 ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this section shall be transferred to a corex glass spectophotometric cell (see 6.4)having a width of approximately 1 centimeter (cm.) and its absorbance density determined at a wave length of 410 millimicrons. As reference, a similar cell containing acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F(A-B)}{w \quad 1-.01M)\ D}$$

Where:

A = optical density of basic lead styphnate solution

B = optical density of ammonium acetate solution

D = width of *corex cell,* cm.

M = moisture content of sample obtained from 4.3.3 in percent

w = weight of sample

F = factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

4.5.4.3 Determination of normal lead styphnate (when applicable). Extract the residue obtained in the barium nitrate determination with sufficient 5 ml. portions of ammonium acetate solution (saturated) at about 25° until the washings are colorless. Agitate each portion for about 1 minute, then allow each portion to remain in contact with the residue for one additional

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minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml. calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml. aliquot of this solution to a second 100 ml. calibrated volumetric flask and “dilute to the mark with distilled water. Transfer a portion of this solution to a *corex* glass spectrophotometric cell having a width of approximately 1 cm. and determine its absorbance density at a wave length of 410 millimicrons . Use as a reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution and measure at a wave length of 410 millimicrons, the absorbance density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

$$\text{Normal lead styphnater percent} = \frac{F\{A-B)}{WD}$$

Where:

A = optical density of normal lead styphnate solution.

B = optical density of ammonium acetate solution.

D = width of corex cell, cm.

w = weight of the dried sample.

F = factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard. .

4.5.4.4 Tetracene. The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60°C. plus or minus 5°C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water,

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and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60°C. plus or minus 5°C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)100}{W\ (1-.01M)}$$

Where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.

D = weight of crucible and residue after treating with hot water.

W = weight of the sample.

M = moisture content of sample obtained from 4.3.3 in percent.

4.5.4.5 Antimony sulfide. The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W\ (1-0.1M)}$$

Where:

D = weight of crucible and sample after treating with hot water .

E = weight of crucible.

W = weight of the sample

M = moisture content of sample obtained from 4.3.3 in percent.

4.5.4.6 Lead azide. The lead azide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

4.5.5 Functioning.

4.5.5.1 Sensitivity. The detonator shall be assembled to the applicable test fixture specified in 4.4.4, so that the firing pin

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will strike the sensitive end of the detonator. A steel ball, approximately 15/16 inch in diameter and weighing 1.94 ounces max imum, shall be dropped on the firing pin, to function the detonator, from a height of 2 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.5.5.2 Waterproofness. The detonator shall be immersed under water as specified for the time specified. Within one hour after removal from the water the detonator shall be subjected to the sensitivity test specified in 4.5.5.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

4.5.6 Heat seal test of sealed bag. Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

5* PACKAGING

5.1 Preservation and Packing.

5.1.1 Level C. Packaging shall be as specified on Dwg. 7548104.

5.1.2 Intermediate packaging. The primers shall be packaged in accordance with dwg. 7548103.

5.2 Packing

5.2.1 Level C. The containers shall be packed in accordance with dwg. 7548099.

5.3 Marking. Inner, outer and exterior containers shall be marked in accordance with dwgs. 7548103, 7548104, 7548099.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, US ARDEC, ATTN: AMSMC-QAT-I (D), Dover, New Jersey 07801-5001.

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6.3 Check test for possible deterioration. The following test is designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in that contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration.

6.3.1 Sensitivity. A sample of 750 detonators shall be selected at random from each lot. If 7 or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.3.5.1 using test equipment as specified in 4.2.4.

6.4 A Quartz ultraviolet spectrophotometer . Described in the Journal of American optical Society, Vol 30 page 683 (1941) was found to be satisfactory for determining the optical density of the styphnate solution.

6.5 Data cards. (See MIL-A-48078). Distribution of data cards shall include the following: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-M , Dover, New Jersey 07801-5001.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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6.8 Subject term (key word) listing.

Detonator, Stab
Loading Assembly
Sensitivity
Waterproofness
Primer
Lead Azide

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A362)

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